



Automated prompt-guided multi-modality cell segmentation with shape-aware classification and boundary-aware SAM adaptation ☆,☆☆

Deboch Eyob Abera ^{a,b}, Jiaye He ^c, Jia Liu ^d, Nazar Zaki ^e, Wenjian Qin ^{a,b,f} ^{*}

^a Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, 1068 Xueyuan Avenue, Shenzhen University Town, Shenzhen, 518055, Guangdong, China

^b University of Chinese Academy of Sciences, No. 19(A) Yuquan Road, Beijing, 100049, Beijing, China

^c School of Pharmaceutical Sciences, Nanjing Tech University, Nanjing, 211816, China

^d National Innovation Center for Advanced Medical Devices, Shenzhen, 518109, China

^e Department of Computer Science and Software Engineering, College of Information Technology, United Arab Emirates University, Al Ain, Abu Dhabi, United Arab Emirates

^f State Key Laboratory of Biomedical Imaging Science and System, 1068 Xueyuan Avenue, Shenzhen University Town, Shenzhen, 518055, Guangdong, China

ARTICLE INFO

Keywords:

Multi-modality cell segmentation
Foundation model
Segment anything model
Classification
Model generalization
Medical imaging

ABSTRACT

Robust and accurate cell segmentation across diverse imaging modalities remains a critical challenge in microscopy image analysis. While foundation models like the Segment Anything Model (SAM) have demonstrated exceptional performance in natural image segmentation, their adaptation to multi-modal cellular analysis is hindered by domain-specific knowledge gaps and morphological complexity. To bridge this gap, we present a novel SAM-driven framework featuring three systematic innovations: First, we propose Shape-Aware Classification to enhance segmentation of cells with diverse morphologies. Second, Auto Point Prompt Generation (APPGen) module guides the segmentation model with automatically generated point cues to improve segmentation accuracy. Third, we implement Boundary-Aware SAM Adaptation to effectively resolve overlapping cells in microscopy images. Our experiments show that the proposed framework reduces manual effort through automated prompts, adapts well to different imaging modalities, and enhances segmentation accuracy by incorporating boundary-aware techniques. The source code is available at https://github.com/MIXAILAB/Multi_Modality_CellSeg.

1. Introduction

Multi-modality cell segmentation refers to the process of accurately identifying and segmenting individual cells from different microscopy modalities, such as Brightfield, Phase Contrast, Differential Interference Contrast (DIC), and Fluorescence microscopy. Together, these modalities enable researchers to extract comprehensive information about cells from organs, advancing our understanding of biological systems and disease mechanisms [1,2].

Cell segmentation methods can be broadly categorized into modality-specific and multi-modality approaches. Modality specific methods are designed and optimized for a particular type of microscopy, such as fluorescence or brightfield, often requiring retraining or fine-tuning when

applied to a different modality [3,4]. In contrast, multi-modality segmentation approaches aim to operate across multiple imaging modalities without significant adaptation [2].

Beyond modality dependence, segmentation approaches can be categorized into three algorithmic paradigms: traditional image processing [5,6], machine learning [7–9], and foundation models [10–12].

Recent advancements in deep learning (DL) across various vision tasks have led to its widespread adoption in microscopy image analysis [13,14]. Notable applications include Omnipose for bacterial cell segmentation [4], Cellpose for general cellular segmentation [15], the 2018 Data Science Bowl challenge for nucleus segmentation across diverse imaging experiments [16], and early work on live cell segmentation [3], all demonstrating strong performance. However, most of

☆ This paper was recommended for publication by Prof. Guangtao Zhai.

☆☆ This work was supported in part by the Ministry of Science and Technology's key research and development program (2023YFF0723400), National Natural Science Foundation of China (62271475), Shenzhen-Hong Kong Joint Lab on Intelligence Computational Analysis for Tumor Imaging (E3G111), Guangdong Youth Talent program (2024TQ08A386), and the Youth Innovation Promotion Association CAS, China (2022365).

* Corresponding author at: Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, 1068 Xueyuan Avenue, Shenzhen University Town, Shenzhen, 518055, Guangdong, China.

E-mail address: wj.qin@siat.ac.cn (W. Qin).

<https://doi.org/10.1016/j.displa.2025.103337>

Received 11 July 2025; Received in revised form 8 December 2025; Accepted 30 December 2025

Available online 7 January 2026

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these methods are designed for specific imaging modalities and often require manual parameter tuning, limiting adaptability and scalability. Developing a universal, fully automated segmentation approach that generalizes across multiple microscopy modalities remains an open and critical challenge.

The Emerging of foundation models has introduced a new paradigm shift at achieving generalization across tasks and modalities. Segment Anything Model (SAM) is one such example that has shown impressive results in natural image segmentation by enabling zero-shot performance through prompt-based guidance [10]. Despite its success in the natural image domain, SAM faces three key limitations when applied to microscopy images. (1) Domain shift — Microscopy images differ significantly from natural images in terms of contrast, resolution, and structural complexity, making accurate segmentation difficult. (2) Prompt dependency — SAM relies on manually provided prompts, limiting full automation and requiring expert input. (3) Instance segmentation limitations — SAM often fails to distinguish individual cells in densely packed or overlapping regions, which is critical for instance-level segmentation in biomedical applications. These limitations underscore the need for task-specific adaptations of foundation models like SAM to address the distinct challenges of multi-modality medical image segmentation.

Recent work on applying SAM to medical image segmentation has advanced in three main areas: semantic segmentation [17,18], interactive prompt enhancement [19,20], and task-specific fine-tuning [21, 22]. In this paper, our aim is to propose a novel approach that addresses the key challenge of cell instance segmentation in multi-modality microscopic images, moving beyond the limitations of designing algorithms tailored to one or a few specific datasets (See Fig. 1). To achieve this, we present four key contributions. First, we introduce the Image Standardization Module, which unifies inputs into a consistent format to address the heterogeneity exhibited in multi-modality microscopy images, such as variations in spatial dimensions, channel configurations, and intensity distributions. This strategy lays the foundation for robust performance in diverse imaging scenarios. Second, we propose Auto-Point Prompt Generation (APPGen) to eliminate the reliance on manual annotations and automate the extraction of segmentation prompts. APPGen leverages distance map guidance and density-based clustering to identify optimal points or regions for segmentation. This strategy simplifies workflows of multi-modality microscopic cell images for large-scale applications. Third, we present the Shape-Aware Classification Module that categorizes objects based on their morphological characteristics, such as round, elongated, tightly packed spherical shapes, irregular forms, and weak boundaries. By incorporating this module, the model gains a deeper understanding of cellular morphology, enabling better convergence and accuracy when trained on diverse cellular structures. Our ablation studies demonstrate that this strategy significantly enhances the model's ability to generalize across varying cell types and arrangements. Finally, we introduce Boundary-Aware SAM Adaptation, a refinement mechanism that leverages distance-map-guided loss and boundary-aware optimization to improve segmentation masks. This strategy specifically addresses the challenge of instance segmentation in overlapping or tightly clustered cells, ensuring precise delineation of individual objects even in complex scenarios. The contributions of this work are summarized as follows:

1. Propose a unified framework to tackle cell instance segmentation challenges across multi-modality microscopic images, ensuring robust performance across diverse datasets.
2. Develop an Auto-Point Prompt Generation (APPGen) module to guide the segmentation model with automatically generated point cues, thereby eliminating the need for manual annotations and improving segmentation accuracy.
3. Propose a Shape-Aware Classification module with boundary-aware SAM Adaptation that uses morphological priors to filter false positives, as well as distance map-guided segmentation to precisely separate overlapping cells, ultimately improving overall accuracy.

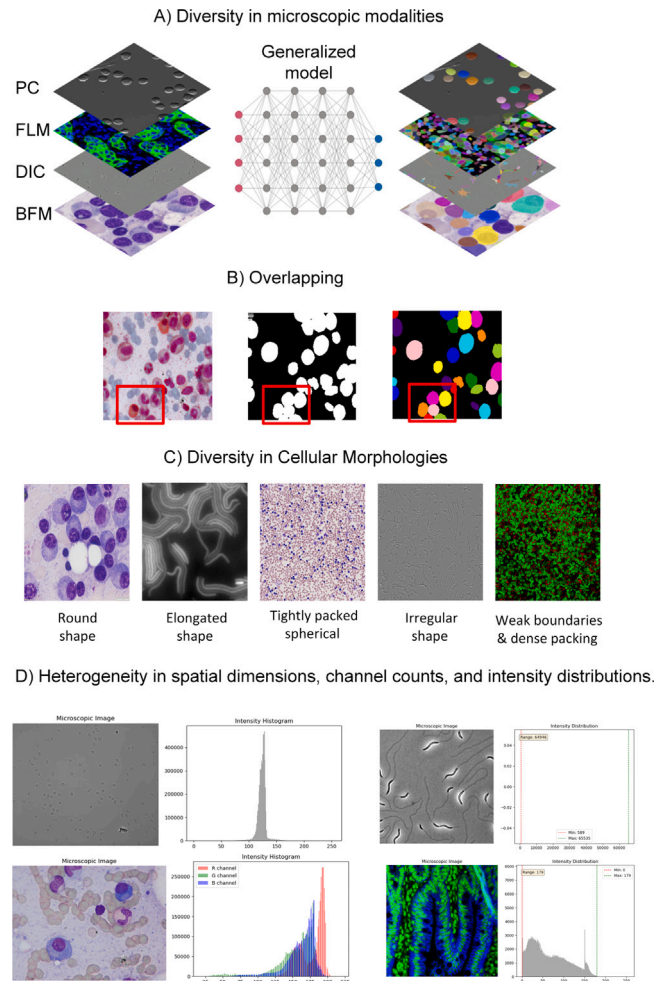


Fig. 1. The Key research challenge in multi-modality cell segmentation: (A) Generalization across diverse microscopy modalities, (B) Overlapping, (C) Morphological variability in cell structures, and (D) Heterogeneity in dimensions, channels, and intensity distributions.

2. Related works

2.1. Base models for segmentation

Previous studies such as Cellpose [15], HoVer-Net [23], StarDist [24] and other methods [1,25,26] have introduced advanced methods to tackle challenges in cell segmentation, particularly in cases involving densely packed or overlapping cells. However, these approaches require extensive manual parameter tuning or retraining when applied to different modality. More recent methods, such as MEDIAR [27], the winner of the Cell Segmentation Challenge, and the SRIBD-Med team [28], the runner-up, focus on improving multi-modality segmentation. However, these models are trained from scratch on base architectures, leading to poor generalization, especially when dealing with imbalanced datasets. Our goal is to develop a generalizable method capable of accurately segmenting cells across diverse microscopy images. This will be our main contribution to this field.

2.2. Foundation models for segmentation

The field of medical image analysis has traditionally relied on specialized neural network architectures such as Convolutional Neural Networks (CNNs) [29], Fully Convolutional Networks (FCNs) [30],

U-Net [31], Generative Adversal Networks (GAN) [32], and more recently, Transformer-based models [33]. These architectures are meticulously designed for the specific modality and objectives in mind, such as segmenting, detecting, or identifying cellular components. To overcome these limitations, Meta AI introduced the Segment Anything Model (SAM) in 2023, a foundation model designed to advance segmentation and enhance generalization across diverse objects in images and videos [10,11]. This shift toward general-purpose foundation models for perception aligns with recent perspectives in biomedical AI, which advocate universal models that bridge diverse data modalities and tasks [34].

Recent efforts to adapt the Segment Anything Model (SAM) for medical image segmentation have primarily focused on semantic segmentation and fine-tuning on specialized datasets [20,35,36]. While this approach succeeds in semantic segmentations, it limits generalization in instance segmentation. This limitation arises because SAM is not designed to handle the unique challenges of overlapping or closely packed structures in cell and nucleus segmentation. As a result, despite fine-tuning on medical datasets, SAM remains limited in its ability to assign unique labels to each instance, particularly when the objects of interest are closely overlapping or touching in the multi-modality microscopic cell images. To address this challenge, we build Boundary-Aware SAM Adaptation with prior shape information, which refines segmentation masks using distance-map-driven fine-tuning.

2.3. Shape-aware and geometry-prior segmentation

Recent advances in biomedical segmentation have emphasized shape priors and geometric constraints to improve instance discrimination in morphologically diverse datasets. Several studies incorporate shape descriptors, contour regularization, or morphological priors to enhance boundary accuracy and structural consistency. Approaches like HoVer-Net [23], StarDist [24], and Cellpose [15] utilize geometric cues and distance embeddings to model nuclei morphology, while others such as KiPA22 Report [37] and Swin-UNet [38] apply contour regularization and attention-based geometric priors to improve segmentation of small or irregular nuclei. In addition, IntegRegion [39] and DeepContour [40] integrate morphological priors and contour-aware losses to maintain object compactness and smooth boundaries. Several recent studies have also explored pseudo-labeling and class-aware embeddings [41, 42] to better manage shape variability, though these approaches can still be affected by noise or misclassification. Despite these advances, most existing models treat morphology implicitly, limiting their ability to generalize across modalities exhibiting high variability in cellular geometry. In contrast, our proposed Shape-Aware Classification explicitly leverages pseudo-labeled morphological classes (round, elongated, irregular, dense, and clustered) and a two-phase hybrid training strategy to guide segmentation with explicit geometric priors. This design improves convergence, strengthens feature discrimination, and mitigates the effects of pseudo-labeling errors—achieving robust, morphology-consistent segmentation across multi-modality microscopy images.

2.4. Boundary-aware instance separation for overlapping cells

Boundary-aware and instance separation strategies have been extensively explored to address the challenge of overlapping and densely packed cells in biomedical images. Traditional approaches often combine distance map learning, boundary loss, and watershed-based post-processing to delineate object borders. Methods such as Cellpose [15], HoVer-Net [23], and StarDist [24] leverage geometric priors or distance-based encodings to enhance instance separation, while marker-controlled watershed and contour refinement techniques further improve boundary localization. However, most existing SAM-based adaptations (e.g., MedSAM [35], MedSAM2D [43]) primarily focus on semantic segmentation without explicitly incorporating boundary cues,

leading to inaccurate instance separation when objects are tightly clustered. To overcome this limitation, our work introduces a Boundary-Aware SAM Adaptation that integrates distance maps into the SAM decoder and jointly optimizes with a hybrid loss (MSE + Gradient Difference Loss). This design enables the model to preserve global object structure while sharpening local edges, learning explicit geometric relationships between neighboring instances. During post-processing, a distance map-guided watershed algorithm refines boundaries and generates precise instance masks (See Fig. 7). This approach bridges the gap between semantic-level SAM adaptations and true instance-level separation, achieving consistent boundary precision and robust segmentation across diverse and densely populated microscopy modalities.

3. Methodology

3.1. Overview of methodology

This section presents an overview of the proposed framework, which consists of two main components – training and testing phases – as illustrated in Fig. 2. The approach is developed to effectively handle the variability of multi-modality microscopy data and ensure precise, adaptive cell segmentation across different imaging platforms and tissue types without additional parameter optimization.

The training process begins with the Image Standardization Module, followed by ground truth refinement, where overlapping cells are separated and distance maps are generated. The third step introduces the Shape-Aware Classification Module, which categorizes objects based on their morphological characteristics (e.g., round, irregular, elongated). This classification step improves model generalization by grouping similar objects and enabling anatomy-specific prompting. It also enhances convergence during training by reducing the complexity of learning from highly heterogeneous data. Next, the Auto-Point Prompt Generation (APPGen) framework is employed to automate the extraction of point prompts for the Segment Anything Model (SAM). This APPGen eliminates the need for manual annotation by leveraging distance maps and density-based clustering to identify representative spatial points. Finally, the training process incorporates Boundary-Aware SAM Adaptation, which optimizes segmentation performance through two complementary loss functions. These loss functions guide SAM to preserve global pixel intensities while sharpening local edges, ensuring precise delineation of cell boundaries and accurate separation of overlapping regions.

During testing, the trained model applies the strategies learned during training to accurately segment cells in unseen multi-modality images. By integrating standardized inputs, the Shape-Aware Classification Module, the Auto-Point Prompt Generation (APPGen) framework, and post-processing of predicted masks, the model achieves consistent and reliable instance segmentation across diverse imaging modalities and tissue types.

3.2. Image standardization module and shape-aware classification

In this section, we discuss the significance of the Image Standardization Module and Shape-Aware Classification for handling multi-modality microscopic images. These components play a critical role in addressing the heterogeneity inherent in diverse microscopy modalities. The complete workflow of this section and a representative sample of classification classes are illustrated in Figs. 3 and 4, respectively.

3.2.1. Image standardization module

Multi-modality microscopy images exhibit significant heterogeneity in spatial dimensions, channel counts, and intensity distributions. To address this, we designed an Image Standardization Module that unifies inputs into a consistent format, enabling seamless integration into downstream tasks. Below are the key steps:

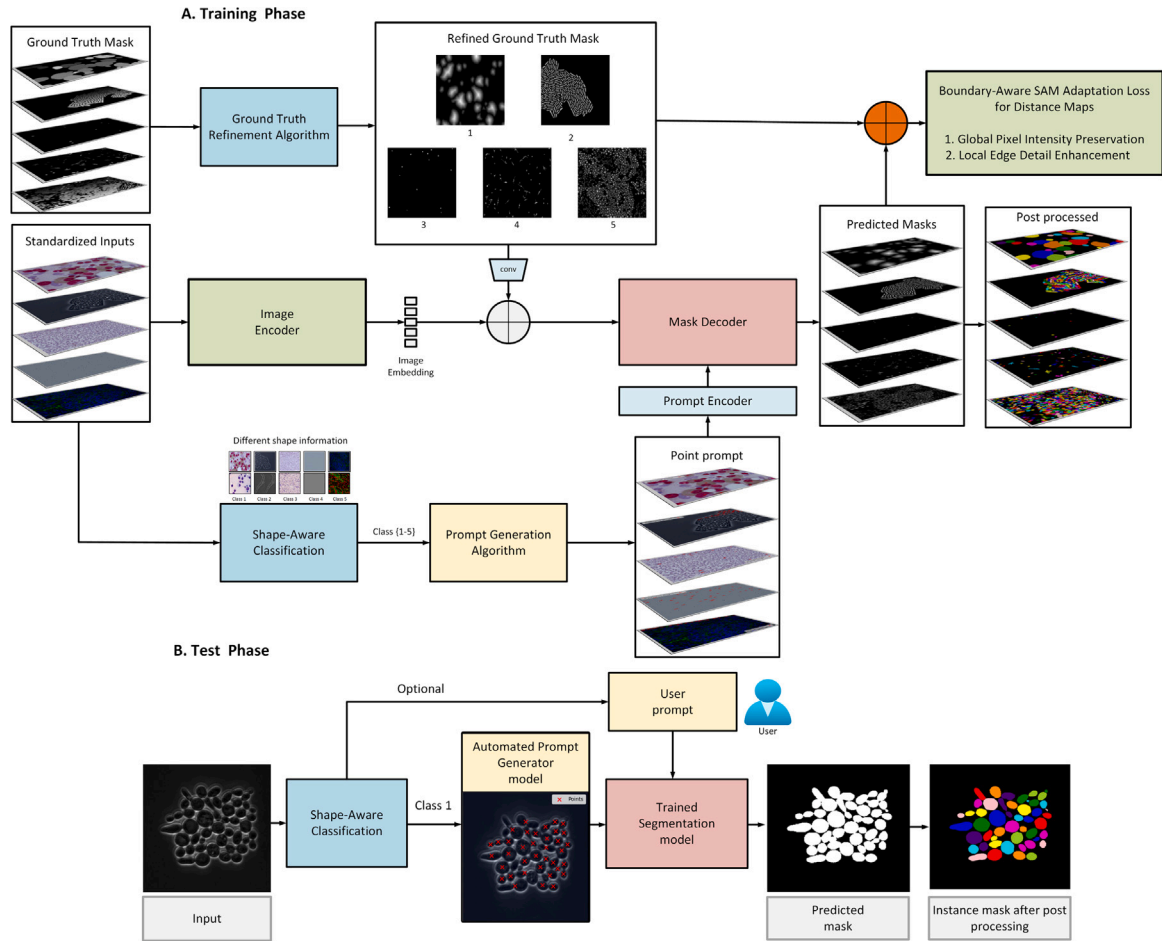


Fig. 2. Overview of methodology. (A) Training phase: The training process begins with refining ground truth to a distance map and multi-modality image standardization, followed by shape-aware classification. Next, auto- point prompts are generated. Finally, incorporates boundary-aware SAM adaptation, which optimizes segmentation performance through global pixel intensity preservation and local edge detail enhancement.(B) Testing phase: Shape-aware classification is followed by automatic or user-provided point prompts, segmentation with the trained model, and instance segmentation via post-processing.

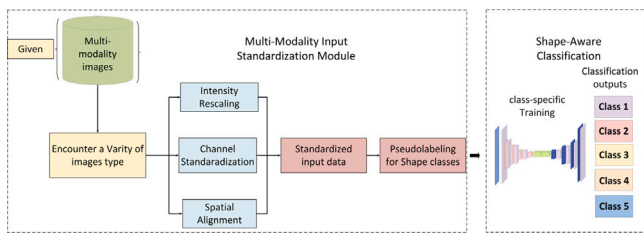


Fig. 3. Pipeline for image standardization module and shape-aware classification.

A. Intensity rescaling: Microscopy images frequently have intensity values outside the standard range of $[0, 255]$ or use non-standard data types (e.g., float or 16-bit integers). To ensure uniformity, we rescale the intensity values of each image to the range $[0, 255]$ using the formula:

$$I_{\text{norm}} = \frac{I - I_{\min}}{I_{\max} - I_{\min}} \times 255 \quad (1)$$

where $I_{\min}$ and $I_{\max}$ are the minimum and maximum pixel values in the input image. This ensures that all images share a consistent intensity range, which is crucial for training deep learning models.

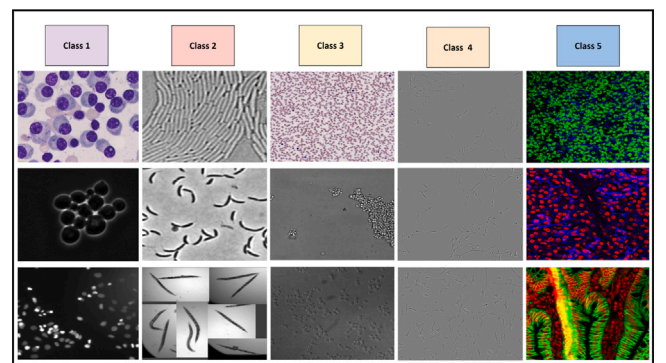


Fig. 4. A representative sample of the classification of pseudo-labeled classes.

B. Channel standardization: Microscopy images may vary in their number of channels (e.g., grayscale, RGB, or multi-channel fluorescence). To standardize the channel configuration: Grayscale images are converted to 3-channel RGB by replicating the single channel across all three dimensions. Images with more than three channels are truncated to retain only the first three channels. Images with fewer than three channels are padded with additional channels derived from the existing

ones. Mathematically, for a grayscale image G of shape (H, W) , the transformation to RGB is given by:

$$G_{\text{RGB}} = \text{stack}(G, G, G) \quad (2)$$

where stack concatenates the grayscale channel along the channel axis.

C. Spatial alignment: Images are resampled to a fixed resolution (e.g., 256×256) using bilinear interpolation to ensure uniform spatial dimensions. This minimizes computational overhead and aligns inputs with pretrained architectures.

3.2.2. Shape-aware classification

In addition to standardizing input images, we introduce Shape-Aware Classification to address the challenges posed by the diverse morphological characteristics of objects in multi-modality images. Objects such as cells and nuclei exhibit a wide variety of shapes, including round, tiny/tightly packed, irregular, elongated, and clustered/densely packed forms (See Fig. 4). Training a model on such heterogeneous data without prior categorization can lead to poor convergence, especially when training from scratch. Even when using foundation models, convergence accuracy is often compromised due to data complexity. To mitigate these issues, we adopt a two-step approach:

A. Pseudo-labeling for shape-aware training: We manually classify the dataset into five classes based on object shape and modality-specific characteristics as shown Fig. 4 (Class 1 “round” cells, Class 2 “elongated” cells, Class 3 “tiny” cells, Class 4 “irregular” cells, and Class 5 “densely packed or clustered boundary cells” cells). This pseudo-labeling step groups similar objects together, simplifying the learning process for the model. The five classes include: Round-shaped objects, elongated objects, small-sized/tightly packed objects, irregularly shaped objects, and densely packed or clustered boundary.

B. Class-specific training. Using the pseudo-labeled dataset, we train a classification model based on ResNet-50. As shown in Fig. 9, the model achieves high accuracy across the confusion matrix and class-wise performance, demonstrating its effectiveness in classifying multi-modality images into predefined categories. This classification step ensures that subsequent tasks, such as segmentation, are performed on well-defined subsets of the data, leading to improved convergence and performance. To evaluate the effectiveness of shape-aware classification, we conducted ablation studies on Table 3 and Fig. 14, comparing models trained with and without this step. The results show that models trained on classified data consistently outperform those trained on unclassified data. Furthermore, to address misclassifications of objects during the pseudo-labeling we adopt a hybrid training strategy: Initially, the model is trained on the entire dataset without classification to ensure broad coverage of all object types. Subsequently, the model is fine-tuned on specific classes to refine its performance for each category. This approach ensures that even if some objects are misclassified during pseudo-labeling, the model still achieves satisfactory results for those cases. The combination of pseudo-labeling, classification, and hybrid training significantly enhances the robustness and accuracy of our framework. This experiment was also included as part of the ablation study, as shown in Table 4.

3.3. Auto-point prompt generation (APPGen)

The Segment Anything Model requires manual input of point prompts to guide segmentation, which limits its automation potential. To address this limitation, we propose the Auto-Point Prompt Generation (APPGen) framework, which automates the extraction of point prompts directly from input images. APPGen leverages distance maps and density-based clustering to identify representative spatial points, enabling SAM to operate without human intervention. Generally, the proposed APPGen model consists of five sequential steps.

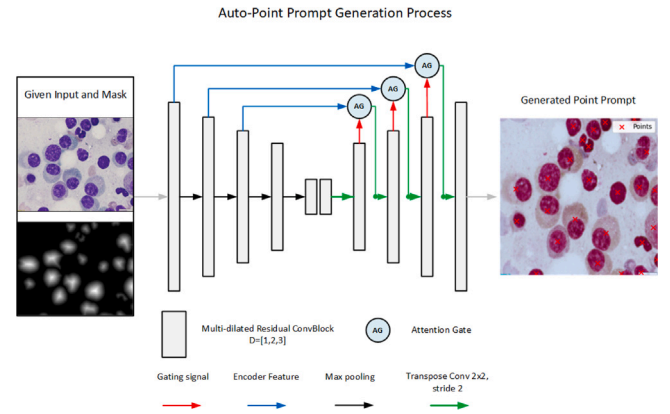


Fig. 5. Processing framework for auto-point prompt generation.

The process begins with model training using the proposed Attention-Guided Multi-Dilated Residual U-Net framework to generate a distance map. Next, the distance map is thresholded at a value of 0.2, converting it into a binary mask. Following this, DBSCAN clustering is applied [44]. In this step, the coordinates of non-zero pixels from the binary mask are extracted and clustered using DBSCAN (Density-Based Spatial Clustering of Applications with Noise). The clustering parameters—neighborhood radius ($\epsilon = 1$) and minimum cluster size ($\text{min_samples} = 5$)—are determined empirically through experiments, as demonstrated in the ablation study (Fig. 17). Clusters containing fewer than five points are considered noise and discarded, ensuring that only dense, meaningful regions are retained.

After clustering, centroid computation is performed. For each valid cluster (excluding noise), the centroid is calculated as the arithmetic mean of its constituent coordinates. This centroid represents the geometric center of the cluster, providing a spatially stable point prompt. Finally, the computed centroids are used as point prompts and fed into SAM to guide segmentation. These automated prompts allow SAM to generate masks without manual intervention, effectively addressing the automation gap in the original framework. As demonstrated in Fig. 5, by combining distance map thresholding, density-based clustering, and centroid selection, APPGen ensures robust and interpretable point prompt generation. The representative results for the APPGen model are further illustrated in Fig. 10, showcasing its effectiveness. This enhancement significantly improves the applicability of SAM in fully automated pipelines, making it a valuable tool for tasks requiring minimal human supervision.

3.4. Boundary-aware SAM adaptation for instance segmentation

The Segment Anything Model (SAM), while effective for general segmentation, struggles in densely packed or overlapping cellular regions due to its lack of explicit boundary awareness. To overcome this, we introduce a Boundary-Aware SAM Adaptation, which incorporates distance maps into SAM’s decoder to enhance boundary precision. The model is trained using distance map supervision and optimized with complementary loss functions that preserve global consistency while sharpening local edges. This adaptation enables accurate separation of overlapping cells across multi-modality microscopy datasets. The overall pipeline of the proposed boundary-aware SAM is shown in Fig. 6.

To prepare the refined ground truth(Distance map) of the model for each instance mask, we compute the Euclidean Distance Transform (EDT). This process is applied to all instances in the image and then combined into a unified distance map. The final combined map is used as the target output during training, allowing the model to learn precise boundaries between objects.

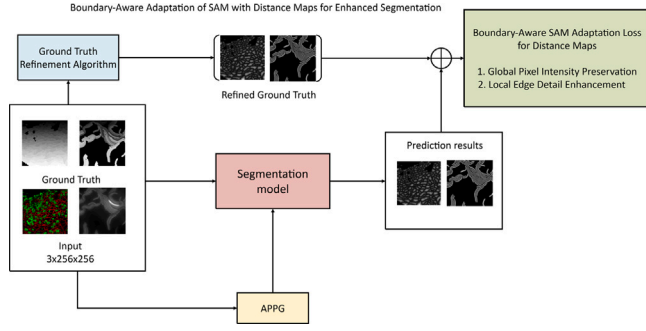


Fig. 6. Pipeline for boundary-aware SAM adaptation in instance segmentation.

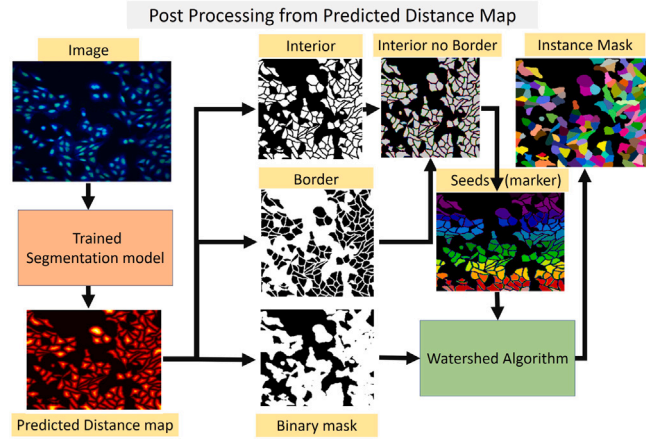


Fig. 7. Proposed post-processing technique from distance map to instance mask.

The training loss for the Boundary-Aware SAM Adaptation model is a combination of two complementary loss functions: Mean Squared Error (MSE) Loss and Gradient Difference Loss (GDL). The Mean Squared Error (MSE) Loss function ensures that the predicted distance map preserves the global structure of the objects, maintaining overall pixel intensity alignment with the target distance map. The MSE loss is formulated as:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{HW} \sum_{x=1}^H \sum_{y=1}^W (D_{\text{pred}}(x, y) - D_{\text{target}}(x, y))^2 \quad (3)$$

where H and W are the spatial dimensions of the distance map, and D_{pred} and D_{target} represent the predicted and target distance maps, respectively. The Gradient Difference Loss (GDL) function enhances the sharpness of object boundaries by penalizing discrepancies in gradient magnitudes between the predicted and target maps. The GDL is computed as:

$$\mathcal{L}_{\text{GDL}} = \frac{1}{HW} \sum_{x,y} \left(|\nabla_x D_{\text{pred}} - \nabla_x D_{\text{target}}| + |\nabla_y D_{\text{pred}} - \nabla_y D_{\text{target}}| \right) \quad (4)$$

where ∇_x and ∇_y represent the horizontal and vertical gradients, computed via finite differences. The total loss is a weighted sum of MSE and GDL:

$$\mathcal{L}_{\text{Total}} = \lambda_1 \mathcal{L}_{\text{MSE}} + \lambda_2 \mathcal{L}_{\text{GDL}} \quad (5)$$

where λ_1 and λ_2 are hyperparameters that control the balance between global structure and boundary sharpness. $\lambda_1 = 0.5$ and $\lambda_2 = 0.6$.

In the post-processing stage (see Fig. 7), the predicted distance map is first thresholded to obtain a binary mask, interior and border

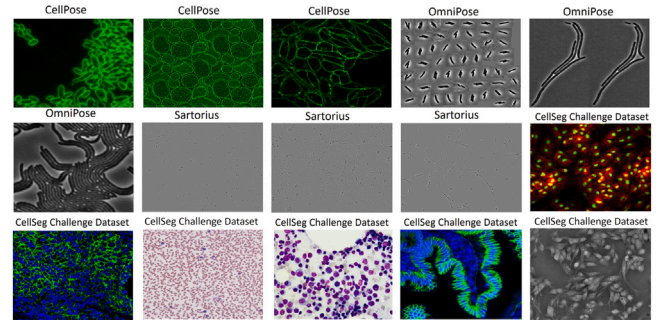


Fig. 8. Representative samples from the public datasets and the CellSeg challenge dataset.

pixels. After removing the border band, the remaining interior regions are converted into seed markers. The watershed algorithm then takes these markers together with the binary mask as input and grows each seed until it meets a border, producing the final instance mask. This is especially helpful in crowded regions, where the default SAM output often fails to separate touching or overlapping cells (see Fig. 21).

4. Experimental results and analysis

4.1. Implementation and training details

Our framework is built upon three core components: a Segmentation model, an Auto-Point Prompt Generative (APPGen) model, and a Classification model. The dataset composition strictly follows the publicly released NeurIPS Multi-modality Cell Segmentation Challenge dataset [2]. The datasets used in our study span both public and challenge-specific resources [2]. The public datasets are OmniPose [4], CellPose [15], and Sartorius [45]. The challenge-specific resources are collected from the NeurIPS Cell Segmentation Challenge. The details of the dataset are described as follows:

- **OmniPose** [4]: collected 735 microscopic images of bacterial cells and worms.
- **CellPose** [15]: collected 608 fluorescence microscopy images of cytoplasm and cellular structures.
- **Sartorius** [45]: collected 606 microscopy images.
- **CellSeg Challenge Dataset** [2]: collected 1000 labeled images, 1500+ unlabeled images, 101 fine-tuning images, and 422 hidden test images of multi-modality microscopy images.

The public datasets allow the model to learn broad morphological and modality variations, while the challenge dataset introduces diverse imaging modalities, tissue types, cell morphologies, and acquisition settings. All datasets provide instance-level cell mask labels, which we convert into distance maps for training. Example images from all sources are presented in Fig. 8.

For segmentation model training, we combined the three public datasets with the 1000 labeled CellSeg images, resulting in 2949 training images, followed by fine-tuning on 101 images, and evaluation on the 422 official test images. For the APPGen model, we applied the same protocol: training on 2949 images, fine-tuning with 101 images, and testing on 422 images. For the classification model, we aggregated all public datasets, the 1000 labeled challenge images, and the 101 fine-tuning images, resulting in 3050 images, split into 80% training and 20% testing. A summary of all datasets, training phases, and test protocols is shown in Table 1.

To ensure experimental reproducibility, all models were trained under clearly defined and consistent conditions. The segmentation framework was initialized from the pre-trained SAM ViT-B checkpoint, and both the encoder and decoder were fine-tuned, allowing the model

Table 1
Summary of framework components, datasets, training, and test protocols.

Component	Datasets used	Training phases	Test phases
Segmentation Model	Public: OmniPose, CellPose, Sartorius. CellSeg Challenge: 1,000 labeled, 101 fine-tuning.	Training: Public + 1,000 labeled. Fine-tuning: 101 fine-tuning images.	✓ 422 CellSeg Challenge
APPGen Model	Public: OmniPose, CellPose, Sartorius. CellSeg Challenge: 1,000 labeled, 101 fine-tuning.	Training: Public + 1,000 labeled. Fine-tuning: 101 fine-tuning images.	✓ 422 CellSeg Challenge
Classification Model	All public datasets + CellSeg Challenge. Split: 80% train, 20% test.	80% train	✓ 20% test

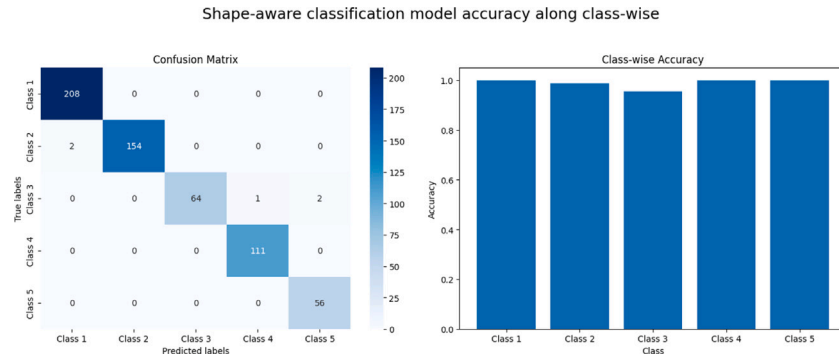


Fig. 9. Shape-aware classification model accuracy along class-wise: The results are presented on a class-by-class basis, providing insights into how effectively the model distinguishes between objects or regions with varying shapes and morphological characteristics.

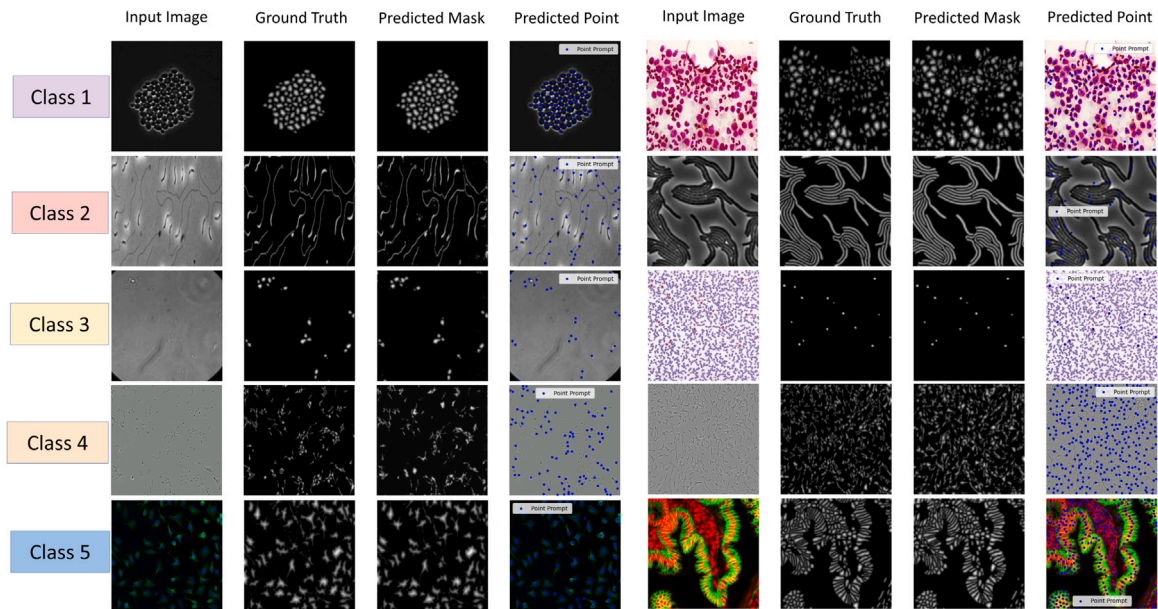


Fig. 10. Auto point prompt generation model performance on different multi-modality data and classes. The results show the APPGen model automatically generates point prompts (e.g., seed points or centroids) to guide segmentation algorithms in identifying regions of interest.

to adapt its image representation and mask decoding modules to the multi-modality training distribution. The initial learning rate was set to 1×10^{-5} and training ran for 100 epochs. The APPGen module, implemented with a U-Net architecture trained from scratch, is optimized using AdamW with a learning rate of 1×10^{-3} for 200 epochs, following the same two-phase protocol as the segmentation model. The classification network is based on a pre-trained ResNet-50, where the final fully connected layer is replaced to support a 5-class morphology categorization task. This model is trained for 100 epochs using AdamW

with an initial learning rate of 1×10^{-4} . All experiments were conducted on an NVIDIA RTX 6000 GPU with 48 GB memory and an Intel Xeon Silver 4210R CPU at 2.40 GHz, using Python 3.9.18 and PyTorch 2.5.0+cu124.

4.2. Evaluation metrics

To evaluate the performance of segmentation models, we employ four evaluation metrics: MSE, intersection over union (IoU), F1 score,

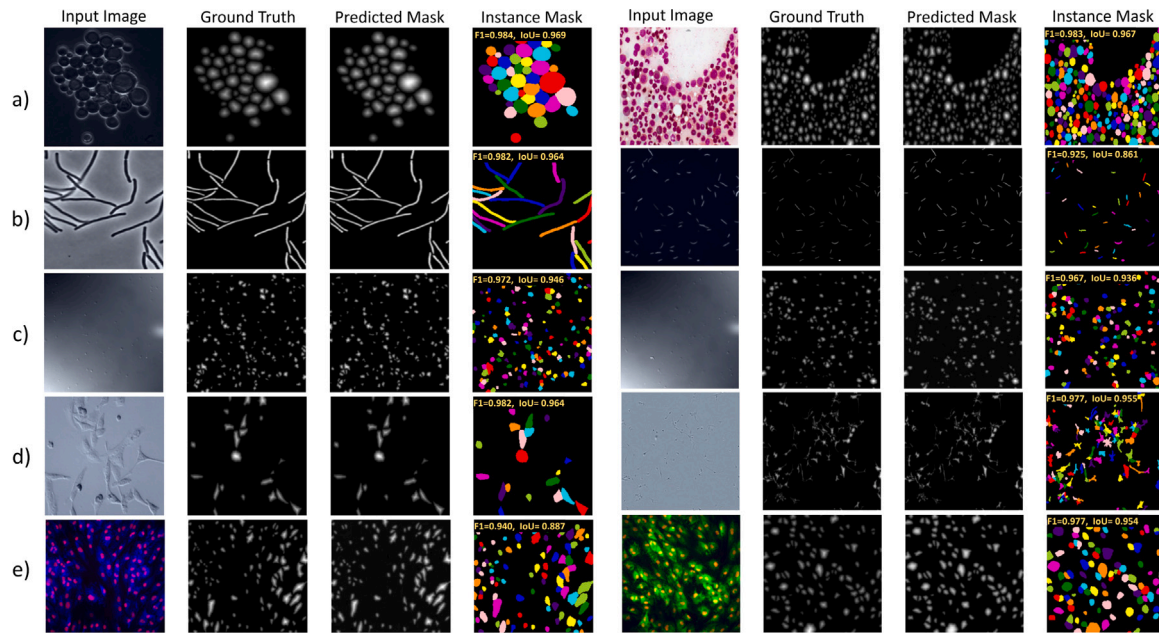


Fig. 11. Proposed segmentation model performance on different multi-modality data and classes. From top to bottom: round, elongated, tightly packed, irregular, and weak boundaries and dense packing. The results highlight the model's ability to handle diverse data types and accurately segment objects or regions with varying characteristics, demonstrating its robustness and versatility in multi-modality settings.

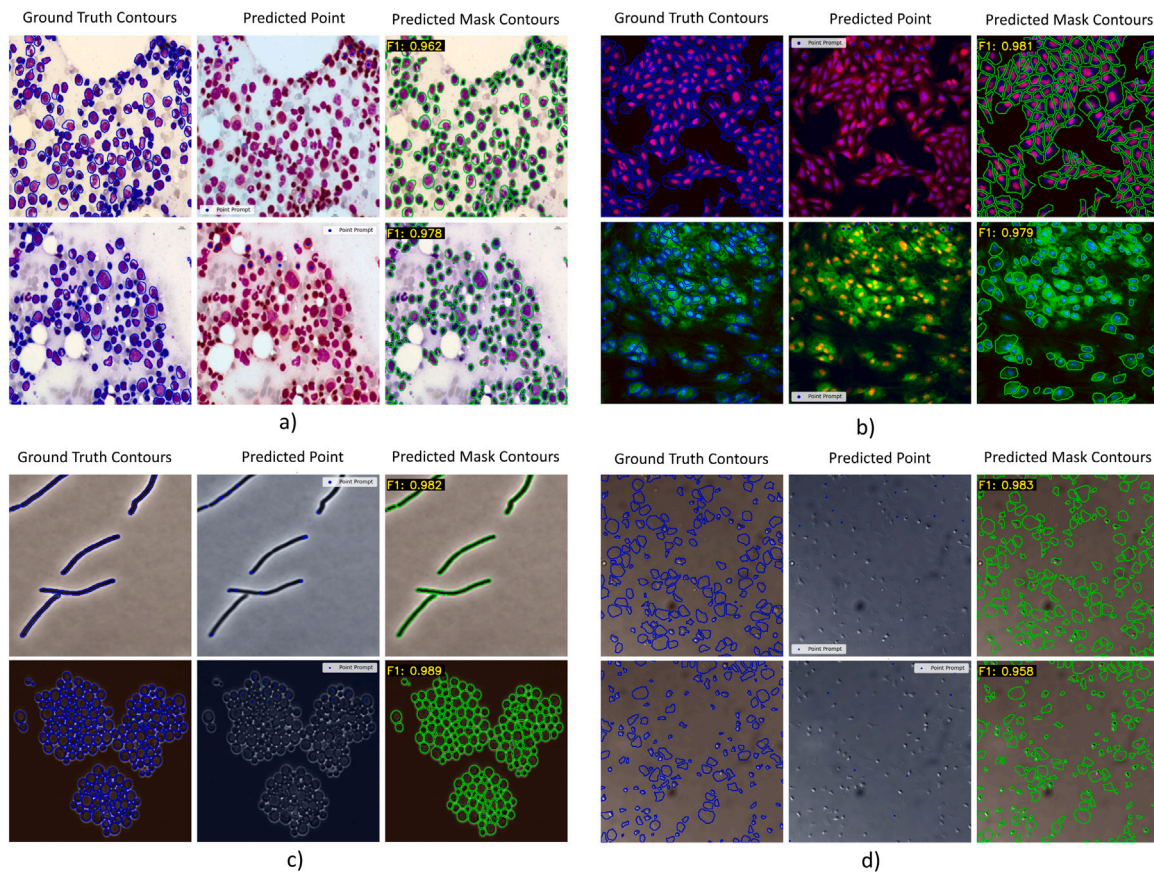


Fig. 12. Segmentation results across multi-modality: (a) Brightfield, (b) Fluorescence, (c) Phase Contrast (PC), and (d) Differential Interference Contrast (DIC). The results show that our model accurately segments cells in diverse modalities.

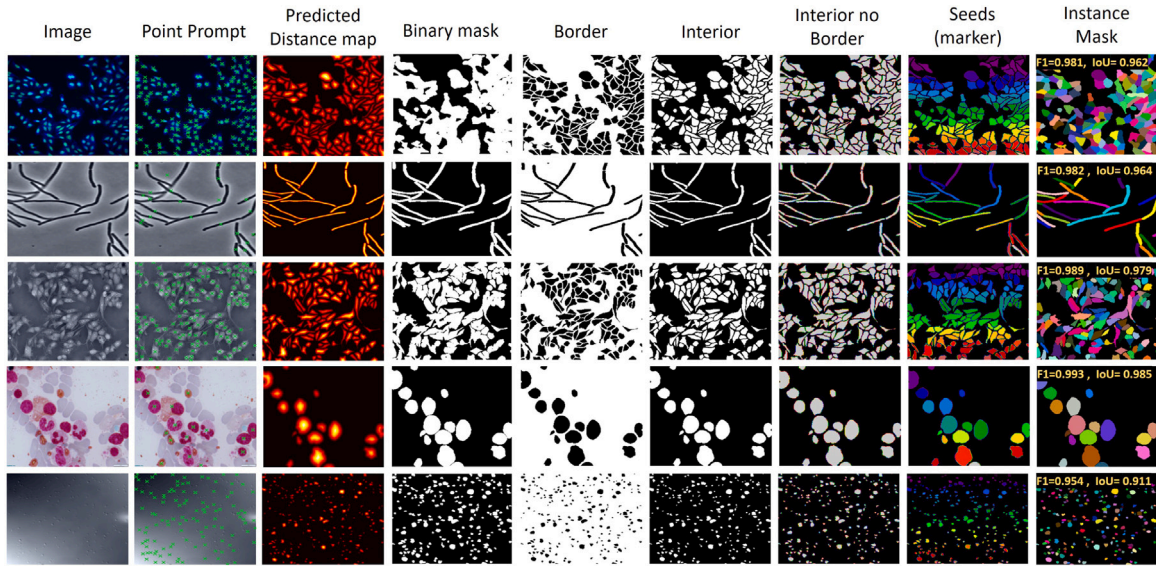


Fig. 13. Representative visual examples demonstrating the complete workflow – from input image and automated point-prompt generation to final instance segmentation – highlighting the model's ability to resolve overlaps accurately.

and panoptic quality (PQ). These metrics provide comprehensive insights into both semantic and instance segmentation tasks. The MSE is a measure of the average squared difference between the predicted and ground truth values, providing a quantitative assessment of the model's accuracy in predicting the distance map. Lower MSE values indicate closer alignment between the predicted and actual distance maps, reflecting better performance in capturing spatial relationships. IoU is another metric used in semantic segmentation, measuring the degree of overlap between the predicted and ground truth masks. The F1 score is a metric used to evaluate the performance of segmentation models by balancing precision and recall. It evaluates whether individual object instances are correctly segmented. PQ is a comprehensive metric used for instance segmentation, assessing both recognition quality (RQ) and segmentation quality (SQ).

4.3. Qualitative experimental result

Our proposed model was trained on OmniPose, CellPose, Sartorius, and CellSeg Challenge datasets. We evaluate the performance of our model on NeurIPS Cell Segmentation Challenge test datasets [2]. Fig. 9 illustrates the performance of the shape-aware classification model in terms of accuracy across different classes. Fig. 10 evaluates the performance of the auto point prompt generation model across various imaging modalities and object classes. The results in Fig. 10 demonstrate the model's ability to generate accurate and meaningful point prompts for segmentation tasks, highlighting its effectiveness in handling diverse data types and class-specific challenges. Fig. 11 presents representative segmentation results from the Challenge datasets. The segmentation outputs in Fig. 11 demonstrate that our model effectively segments cells in different morphological classes without any noticeable performance degradation. Fig. 12 illustrates segmentation results across multiple microscopy modalities – brightfield, fluorescence, phase contrast (PC), and differential interference contrast (DIC) – demonstrating the model's robustness in densely packed and heterogeneous cell environments. Finally, Fig. 13 presents visual examples of the end-to-end workflow, showing the process from input images and automated point-prompt generation to post-processed instance segmentation, highlighting the model's effectiveness in accurately resolving overlapping cells. Overall, our model successfully addresses the key challenges in multi-modality cell microscopy imaging which outlined in Fig. 1 of the Introduction. 1, achieving consistent segmentation performance across diverse modalities and complex cellular structures.

Table 2

Performance metrics for different modalities.

Modality	Images	Cells	MSE	IoU	F1	PQ
Fluorescence	122	12 900	0.10	91.45	97.44	92.77
Brightfield	120	17 880	0.08	92.24	98.59	95.65
Phase Contrast (PC)	120	25 004	0.14	93.11	97.05	89.36
Differential Interference Contrast (DIC)	60	9103	0.06	91.70	97.26	87.72
All Test Images	422	64 887	0.10	92.18	97.63	91.90

4.4. Quantitative experimental result

To evaluate the performance of the segmentation model, we employed four widely recognized metrics: MSE, Intersection over Union (IoU), F1 score, and Panoptic Quality (PQ). The evaluation was conducted using 422 test images from the Cell Segmentation Challenge, which comprises multi-modality microscopic images, including 120 Bright Field images with 17,880 cells, 122 Fluorescent images with 12,900 cells, 120 Phase Contrast (PC) images with 25,004 cells, and 60 Differential Interference Contrast (DIC) images with 9103 cells. The primary objective of this challenge is to facilitate the development of universal cell segmentation algorithms capable of accurately segmenting a wide range of microscopy images without requiring manual intervention. In Table 2, the model's performance across the four evaluation metrics is presented, demonstrating satisfactory results across all metrics. Lower MSE values indicate closer alignment between the predicted and actual distance maps. Similarly, a high Intersection Over Union (IoU) score demonstrates the model's effectiveness in minimizing false positives and false negatives, ensuring precise overlap between the predicted and actual regions. Furthermore, the high F1 score highlights the model's balanced performance in terms of precision and recall, underscoring its robustness in handling diverse cell morphologies. Moreover, the high Panoptic Quality (PQ) signifies the model's capability to achieve both accurate semantic segmentation and instance separation, making it particularly effective for multi-modality datasets. These results collectively validate the efficacy of our proposed approach in addressing the challenges posed by complex microscopy images, demonstrating its potential as a reliable tool for universal cell segmentation tasks.

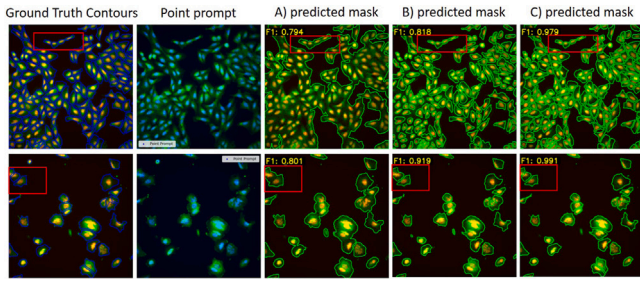


Fig. 14. Ablation study comparing segmentation approaches. From left to right: ground truth instances; selected five-point prompts; segmentation trained with semantic masks; segmentation trained with separated distance maps; segmentation trained with distance maps and shape-aware classification. Results demonstrate that training with distance maps significantly improves accuracy over semantic masks alone. Additionally, the best result achieved training using the shape-aware classification strategy.

Table 3

Ablation study on model training strategies evaluated using a 400-sample hidden test set. Shape-Aware: model trained on shape-aware classification; Semantic-Mask: training on the semantic mask; Boundary-Aware: training boundary-aware SAM with distance map and optimizes segmentation performance through two complementary loss functions.

Shape-Aware	Semantic-Mask	Boundary-Aware	F1	IoU
×	✓	×	0.7923	0.6856
×	×	✓	0.8180	0.7217
✓	×	✓	0.9758	0.9549

Table 4

Morphology classes and their associated F_1 scores on a hidden test set of 400 samples. The dataset distribution includes: Class 0 (185 images), Class 1 (30 images), Class 2 (66 images), Class 3 (31 images), and Class 4 (88 images). Columns F_1^1 , F_1^2 , F_1^3 , F_1^4 , and F_1^5 represent the segmentation results for each class, with corresponding F_1 scores indicating model performance.

Morphology shapes	Class	F_1^1	F_1^2	F_1^3	F_1^4	F_1^5
Round	1	98.33	95.64	94.25	91.42	91.89
Elongated	2	88.67	98.45	93.18	88.67	89.85
Tightly packed spherical	3	35.81	38.53	97.05	88.79	84.75
Irregular	4	70.85	70.42	69.15	93.88	84.98
Weak boundaries and dense packing	5	56.43	59.21	61.97	60.35	97.44
Given all multi-modality	1-5	75.90	76.43	85.54	83.88	91.25

Table 5

Quantitative performance of the proposed model across different shape classes: Class 1 (round cells), Class 2 (elongated cells), Class 3 (tiny/tightly packed cells), Class 4 (irregular cells), and Class 5 (densely packed or clustered boundary cells).

Class	#Images	IoU (↑)	F1 (↑)	MSE (↓)
Class_1	185	0.9684 ± 0.0444	0.9833 ± 0.0269	0.0009 ± 0.0027
Class_2	30	0.9696 ± 0.0125	0.9845 ± 0.0065	0.0008 ± 0.0021
Class_3	66	0.9430 ± 0.0256	0.9705 ± 0.0140	0.0008 ± 0.0022
Class_4	31	0.8949 ± 0.1232	0.9388 ± 0.0917	0.0011 ± 0.0041
Class_5	88	0.9505 ± 0.0258	0.9744 ± 0.0139	0.0010 ± 0.0019
Overall	400	0.9549 ± 0.0514	0.9761 ± 0.0338	0.0009 ± 0.0026

4.5. Ablation study

Our proposed framework is built upon the Segment Anything Model (SAM), which we fine-tuned across the encoder, decoder, and prompt encoder to adapt it for complex biomedical segmentation tasks. To enhance SAM's capability in handling overlapping, heterogeneous, and morphologically diverse cells, we introduced three core innovations: automatic point-prompt generation, distance-map-guided training, and

shape-aware feature integration. In the ablation study, we systematically analyzed the significance of each innovation: First, we demonstrated how distance-map-guided training effectively resolves the overlapping problem by providing spatial and boundary cues that help the model distinguish touching nuclei. Second, we showed the contribution of shape-aware information, which refines contour precision and reduces boundary ambiguity, thereby improving segmentation accuracy. Third, we evaluated the automatic point-prompt generation mechanism, including the parameter selection criteria for the DBSCAN-based centroid extraction, confirming its robustness and consistency across cell types and densities. In addition, we analyzed representative misclassification scenarios to highlight how each module improves the network's interpretability and generalization.

Table 3 reports the ablation study on a 400-sample hidden test set, comparing (i) training with semantic masks only, (ii) training with distance maps, and (iii) addition of shape-aware classification. The results demonstrate that distance-map supervision significantly enhances segmentation accuracy over semantic training alone, and the best performance is achieved when the shape-aware module is included. Fig. 14 visually supports these findings, showing how boundary-aware and shape-aware strategies collectively improve accuracy and instance separation. To further contextualize our improvements, Fig. 15 provides a direct comparison with state-of-the-art SAM-based models in medical imaging, including MedSAM [35] and MedSAM2D [43]. Unlike these models – which rely on user-provided prompts and semantic maps – our method automatically generates point prompts and leverages learned geometric priors. As shown in Fig. 15, our model consistently outperforms both MedSAM and MedSAM2D, demonstrating clear advantages in segmentation accuracy, robustness, and practical applicability.

In Table 4, we further analyze the model's performance across individual classes to evaluate accuracy for different cell morphologies. A key objective of our study is to evaluate the stability of the framework under potential misclassification by the shape-aware module. In Table 4, each row represents the true cell morphology, and each column (F_1^1 , F_1^2 , F_1^3 , F_1^4 , F_1^5) represents the segmentation performance when that morphology is processed by Model 1 through Model 5. The results demonstrate that each model performs best on its own target morphology, confirming successful specialization. Performance decreases when a morphology is handled by the wrong model, indicating that classification errors do influence segmentation; however, the degradation remains moderate, showing that the pipeline is resilient even in misclassification scenarios. This analysis verifies that the system maintains reliable segmentation behavior even when the routing model makes occasional mistakes. This robustness is supported by our hierarchical two-phase training strategy. In the first phase, all models are trained on the full multi-modality dataset to learn broad morphological priors. In the second phase, each model is fine-tuned on its own morphology category, correcting class-dependent bias and reinforcing geometric specialization. This hybrid optimization – combined with boundary-aware supervision – substantially reduces the impact of noisy pseudo-labels and enhances consistency across all morphological types. As shown in Fig. 16, without this second training stage, the model sometimes misinterprets internal cellular particles – especially in densely packed Class 5 samples – as separate target objects, which in turn causes boundary omissions and inaccurate instance separation. After hybrid fine-tuning, these errors are eliminated, and consistent segmentation is achieved across all classes. Importantly, our classification accuracy exceeds 90% for all morphological categories (see Fig. 4 and Fig. 9), confirming that pseudo-label-induced misclassification is rare and has negligible impact on overall segmentation quality.

Fig. 17 presents a systematic sensitivity analysis for automatic prompt generation. We initially applied a grid search (threshold = 0.2, $\epsilon = 2$, min_samples = 5), and through systematic tuning, we discovered that threshold = 0.2, $\epsilon = 1$, min_samples = 5 provides the optimal configuration. Importantly, performance remained stable across nearby parameter settings, confirming the robustness and reliability of our

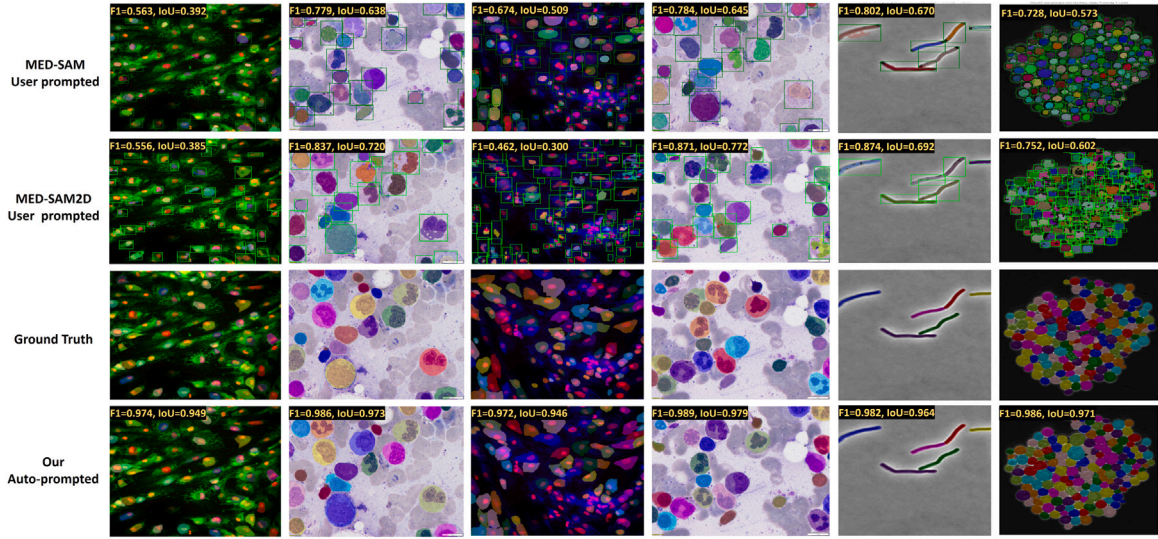


Fig. 15. Performance comparison with SAM-based baselines. From top to bottom: Med-SAM (user-prompted), Med-SAM2D (user-prompted), ground truth, and our proposed (auto-prompted). Our method achieves the highest accuracy on the multi-modality test set, outperforming all baseline models.

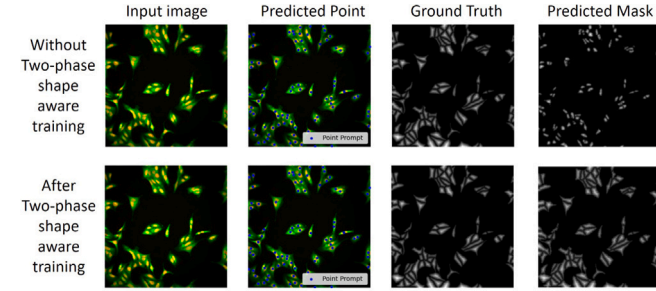


Fig. 16. Effect of two-phase shape-aware training on dense cell segmentation. Without the hybrid two-step training, the model tends to misinterpret internal cell components as separate nuclei in densely packed cells.

prompt generation mechanism. Furthermore, in this ablation section Fig. 18, we demonstrate that the model does not require one prompt per cell – even when processing thousands of nuclei. As long as the target objects share similar morphology, the model can accurately segment multiple instances using only a few representative prompts, demonstrating strong efficiency and generalization. As shown in Fig. 10, our model reliably identifies and generates prompts across different regions, while Fig. 18 confirms that accurate segmentation is maintained even with minimal prompting. To strengthen statistical validity, we include mean $\pm$ standard deviation metrics in Table 5, confirming that the observed improvements are consistent and reproducible. These results demonstrate that the model maintains stable accuracy across all morphological classes and modalities, ensuring that the reported gains are robust rather than incidental.

4.6. Comparison with existing methods

To demonstrate the performance of our model, we compared it with state-of-the-art (SOTA) methods in cell segmentation, including MedSAM [35], MedSAM2D [43], Cellpose [15], and Omnipose [4], as well as the top-performing algorithm KIT-GE [46] from the CTC cell segmentation task [47]. Additionally, among the 28 algorithms that participated in the multi-modality cell segmentation challenge [2], we selected the top three methods: MEDIAR [27], the SRIBD-Med team [28], and T3-cells [48]. We trained KIT-GE, Omnipose and Cellpose on the challenge's training set and then used them to predict test

images. Notably, these algorithms share a common foundation, as they are all constructed on CNN architectures, such as U-Net. For the top three methods (MEDIAR [27], SRIBD-Med team [28], T3-cells [48]), MedSAM [35], and MedSAM2D [43], we utilized the pre-trained models available on their provided GitHub repositories. Fig. 19 provides a qualitative comparison across a wide range of modalities – including fluorescence, brightfield, phase-contrast, and DIC – against eight state-of-the-art (SOTA) segmentation frameworks. Fig. 20 further quantifies this comparison using F1 scores across modalities. Additionally, Fig. 2 highlights performance specifically on overlapping and densely packed cell structures, which represent the most challenging cases in multi-modality cell segmentation.

Traditional instance-segmentation pipelines such as CellPose [15], Omnipose [4], and KIT-GE [46] exhibit strong performance when evaluated on their designated datasets. However, they break down under cross-modality testing. These architectures inherently encode modality-specific assumptions and therefore struggle when large variations in texture, staining, illumination, or contrast appear. In our experiments, these models frequently produce unstable masks, fragmented boundaries, and significant drift due to the domain shift across modalities. Their performance degrades sharply on fluorescence and DIC images, showing divergence when applied to large, diverse test sets with a single unified model. Models built on prompt-driven vision foundation architectures, such as MedSAM and MedSAM2D, offer improved structural consistency but are heavily dependent on user-provided prompts. Without manual point guidance, they produce incomplete segmentation and fail to resolve overlapping regions. In contrast, our method introduces automatic prompt generation, enabling the prompt encoder to function without any user input. As demonstrated in Fig. 15 19, 21, this eliminates manual intervention and significantly improves the accuracy and stability of the predicted instances. Advanced deep architectures such as MEDIAR [27], the SRIBD-Med Team [28], and T3-Cells [48] achieve competitive results on specific modalities but suffer from generalization gaps. Their performance is sensitive to imbalanced training distributions and tends to collapse on minority modalities. These methods inherit capacity limitations from their backbone networks, which are insufficient to model the extreme variability present across microscopy styles. Consequently, they exhibit: inconsistencies across modalities, failure modes on faint or low-contrast cells, and limited robustness to overlapping and tightly clustered regions (See Fig. 21). Our proposed method resolves these limitations by leveraging a foundation-model-based encoder with distance-map guidance, automatic point-prompt generation, and shape-aware regularization. This

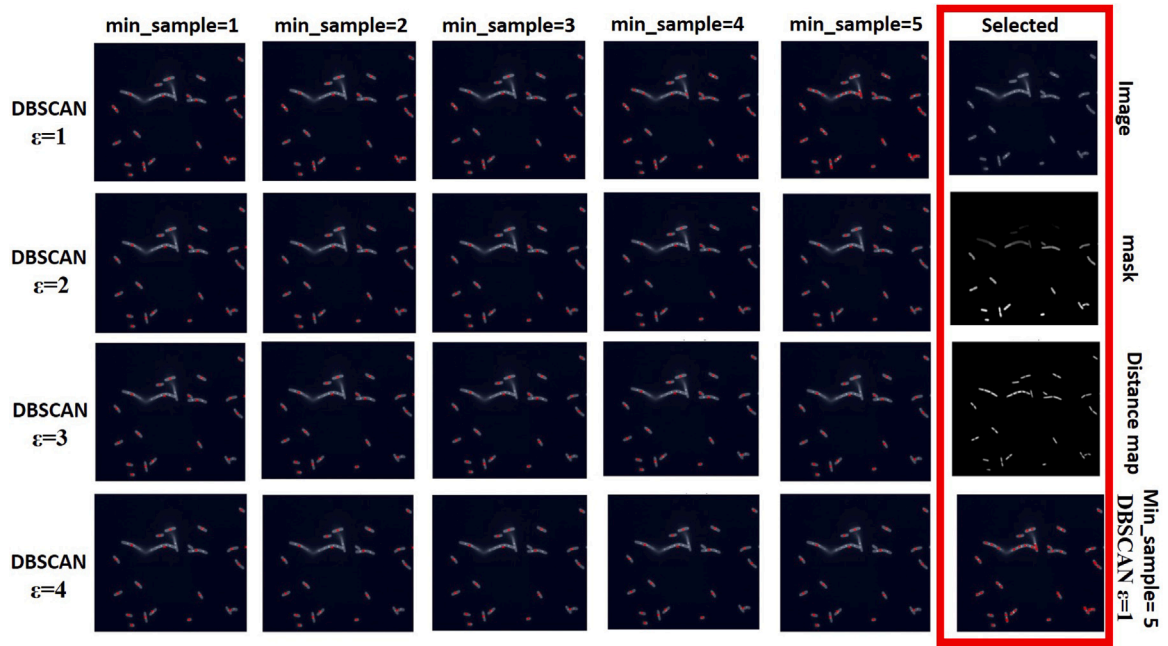


Fig. 17. Parameter sensitivity analysis for the Automatic Prompt Generation (APPGen) module. The robustness of APPGen is evaluated by varying its key parameters: distance map threshold (0.1–0.5, with 0.2 selected), DBSCAN clustering radius ϵ (1–4), and minimum cluster size $min_samples$ (1–5). The plots show that point prompt generation performance remains high and stable across a wide range of values, with only minor degradation at extreme settings (e.g., large ϵ causing multiple cell centers to merge). The optimal configuration (threshold = 0.2, ϵ = 1, $min_samples$ = 5) achieves the best balance between precision and robustness.

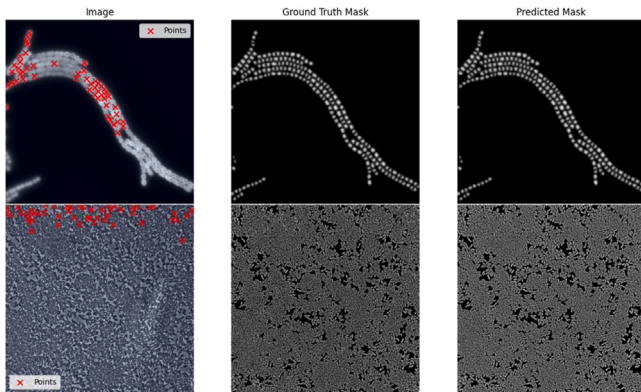


Fig. 18. Segmentation with a limited number of prompts. Even when provided with only a few representative point prompts (far fewer than one per cell), the proposed method produces accurate and consistent cell segmentations. This shows that our model's segmentation masks closely match the ground truth, confirming that the model does not require a prompt for every cell and can generalize from minimal prompt input.

combination provides: Stable convergence across minority modalities, High fidelity boundary recovery even in densely packed regions, and Superior generalization under multi-modality domain shifts. As shown in Fig. 19, and Fig. 20 (quantitative bar plot), our model achieves the highest F1 score across all modalities, outperforming all eight competing methods by a clear margin. Moreover, Fig. 21 illustrates its strong capability in overlap resolution, where competing methods either merge adjacent cells or introduce mask fragmentation. Overall, these results confirm that the proposed framework provides state-of-the-art multi-modality generalization, robust overlapping handling, and fully automated prompt-free instance segmentation, positioning it as a substantial improvement over existing SOTA solutions.

5. Conclusion

In this study, we proposed a universal cell segmentation framework designed to accurately segment diverse cell morphologies across multiple microscopy modalities, including bright-field, fluorescence, phase-contrast, and DIC imaging. The framework integrates four key components: an image standardization module for modality-invariant processing, an automated point prompt generation mechanism that eliminates the need for manual interaction, a shape-aware classification to improve accuracy in complex morphologies, and boundary-aware adaptation of the Segment Anything Model (SAM) to better resolve overlapping cells. Our method demonstrates strong performance across a wide range of datasets, achieving the highest results among the 28 participating algorithms in the 2022 NeurIPS Cell Segmentation Challenge, both per modality and overall. Compared to the current state-of-the-art method, MEDIAR, our approach improved the F1 score by an average margin of approximately 0.12, achieving an overall F1 score of 0.976 versus MEDIAR's 0.857, with consistently higher scores in all individual modalities. Compared to other leading methods such as Cellpose, Omnipose, and KIT-GE, our model achieved an average F1 score of 0.976, significantly outperforming Cellpose (0.569), Omnipose (0.415), and KIT-GE (0.374). This improvement of approximately 40–60 percentage points highlights the superior accuracy, robustness, and generalization capabilities of our proposed approach.

Although our model generalizes well across common modalities and cell types, its performance may degrade when applied to rare or unseen morphologies without fine-tuning. Additionally, the current implementation is limited to 2D images; extending the framework to 3D volumetric data would further broaden its applicability in advanced biomedical applications.

This work contributes to the field of biomedical image analysis by offering a flexible, accurate, and largely automated segmentation pipeline. By reducing reliance on manual annotation, our approach supports high-throughput analysis and improves reproducibility in cellular studies. Moreover, the successful adaptation of SAM to microscopy

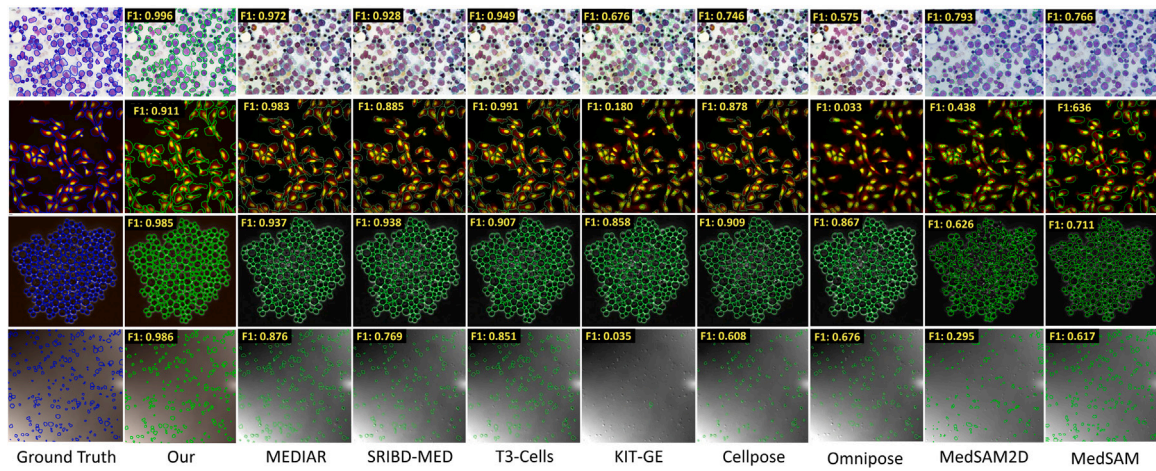


Fig. 19. Comparison of different segmentation methods: The figure presents the segmentation performance of various methods, evaluated using F1 scores. From top to bottom, the modalities include Brightfield, Fluorescence, Phase Contrast (PC), and Differential Interference Contrast (DIC). Our proposed model demonstrates accurate instance segmentation across multiple modalities.

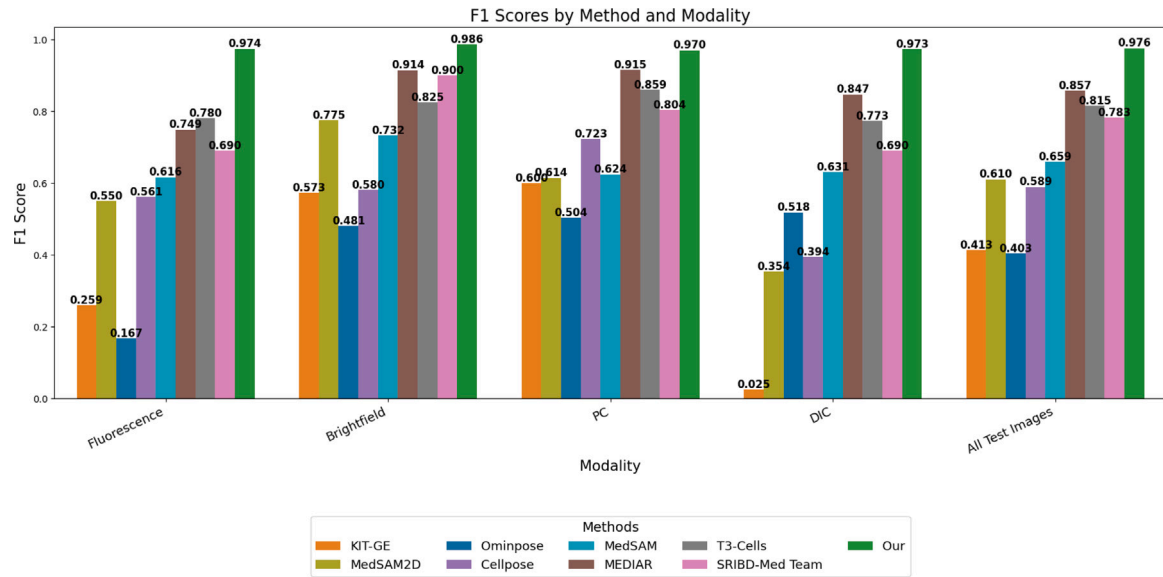


Fig. 20. Comparison of performance across different methods and modalities: The visualization illustrates the F1 scores of various segmentation methods evaluated on different imaging modalities, including Fluorescence, Brightfield, Phase Contrast (PC), and Differential Interference Contrast (DIC). Additionally, overall performance metrics on all test images are provided to offer a comprehensive assessment of each method's robustness and generalization across diverse image types and cellular contexts.

data opens new avenues for leveraging foundation models in domain-specific tasks. Looking ahead, we plan to extend the framework to support 3D and time-lapse segmentation, enabling dynamic tracking of cellular behavior across modalities and organs. This will be particularly valuable in regenerative medicine and developmental biology, where understanding tissue and organ development at single-cell resolution is critical. We also intend to explore semi-supervised learning strategies to reduce dependency on large annotated training datasets, further enhancing the model's practical utility.

CRediT authorship contribution statement

Deboch Eyob Abera: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Jiaye He:** Writing – review & editing, Investigation. **Jia Liu:** Writing – review & editing, Investigation. **Nazar Zaki:** Writing

– review & editing, Investigation, Validation. **Wenjian Qin:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization..

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

I have shared the public dataset link.

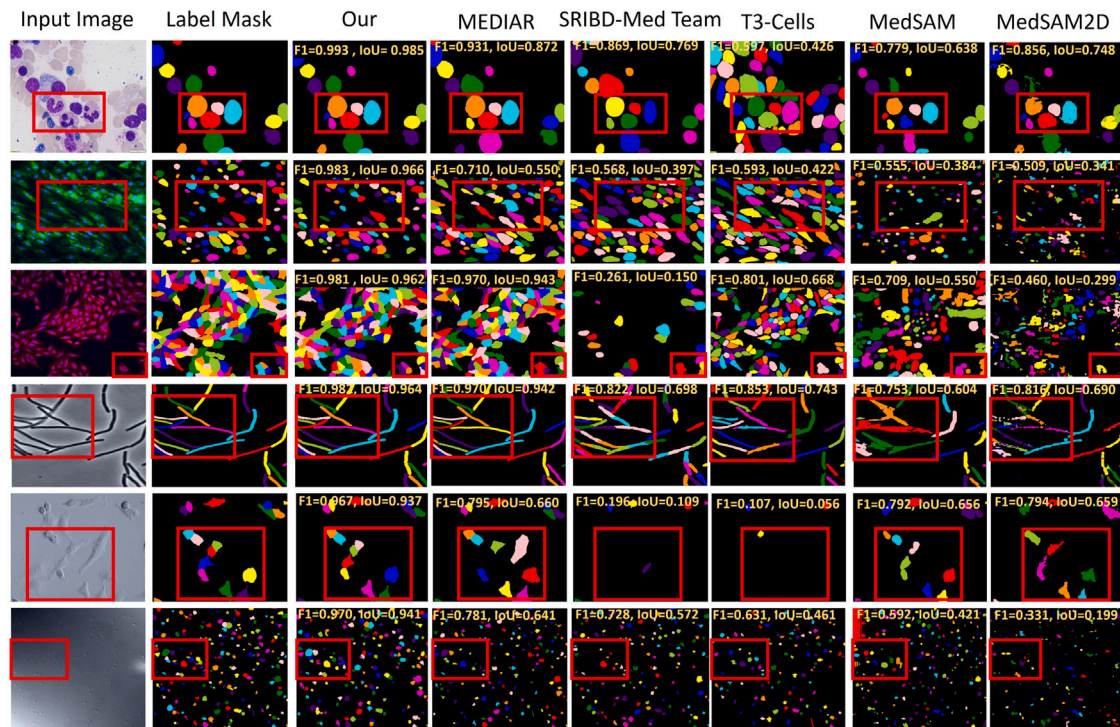


Fig. 21. Comparative results against several state-of-the-art models, including MedSAM, MEDIAR, MedSAM2D, T-3 Cells, and SRIBD-Med, for evaluating the handling of overlapping cells. The results demonstrate that our model more effectively resolves the overlapping cell problem compared to other methods.

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